10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 8.1 — Chords, Distance from the Centre and Radius

Unit 8 | Solved Exercise | New Book (PECTAA)

Exercise 8.1 (Solved)

Two facts do almost all the work here. (1) The perpendicular drawn from the centre to a chord bisects the chord. (2) Chords that are equal in length are equally far from the centre, and chords equally far from the centre are equal.

1. Calculate the length of a chord which stands at a distance of 5 cm from the centre of a circle whose radius is 13 cm.

Let AB be the chord, M its midpoint and O the centre

$OM \perp AB$, $OM = d = 5$ cm, $OA = r = 13$ cm

In the right triangle OMA

$AM^2 = OA^2 - OM^2$

$AM^2 = 13^2 - 5^2 = 169 - 25 = 144$

$AM = 12$ cm

The perpendicular bisects the chord, so $AB = 2 \times AM$

$AB = 2 \times 12 = 24$ cm

Answer: Length of the chord = 24 cm

2. In construction, three steel rods are fixed at points A, B and C (not in a straight line). A circular hoop needs to pass through all three. How many hoops can be used?

Three non-collinear points are given

Through three non-collinear points one and only one circle can pass

Answer: Only one hoop can pass through all three rods

3. In a park, lamp posts are 14 m apart on the edge of a circular part of radius 10 m. Find the distance of the chord from the centre of the park.

Chord AB = 14 m, $r = OA = 10$ m, M is the midpoint of AB

$AM = \frac{14}{2} = 7$ m

In the right triangle OMA

$OM^2 = OA^2 - AM^2$

$OM^2 = 10^2 - 7^2 = 100 - 49 = 51$

$OM = \sqrt{51} = 7.14$ m

Answer: Distance of the chord from the centre ≈ 7.14 m

4. In a circle, chords AB and CD both have length 10 cm. If the distance from the centre to AB is 6 cm, what is the distance from the centre to CD?

Equal chords of a circle are equidistant from the centre

$AB = CD = 10$ cm

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$\therefore$ distance of CD from the centre = distance of AB from the centre

Answer: Distance of CD from the centre = 6 cm

5. Two holes A and B are drilled 12 cm apart on a circular tabletop of radius 10 cm. Find the perpendicular distance from the centre to AB.

Chord AB = 12 cm, r = 10 cm, AM = $\frac{12}{2}$ = 6 cm

$OM^2 = OA^2 - AM^2 = 10^2 - 6^2 = 100 - 36 = 64$

OM = 8 cm

Answer: Perpendicular distance from the centre = 8 cm

6. A chord 8 cm long is at a distance of 3 cm from the centre. Calculate the radius of the circle.

Half the chord = $\frac{8}{2}$ = 4 cm, d = 3 cm

$r^2 = d^2 + (\text{half chord})^2$

$r^2 = 3^2 + 4^2 = 9 + 16 = 25$

r = 5 cm

Answer: Radius of the circle = 5 cm

7. In the given figure, mCD = 12 units and mCB = 3x units. Find the value of x.

In the figure OB is perpendicular to the chord CD, so the radius bisects the chord and the arc CB is half of the arc CD.

$OB \perp CD \implies CB = \frac{1}{2}$ CD

$3x = \frac{1}{2} \times 12$

3x = 6

x = 2

Answer: x = 2

8. In a circle with centre at O, the perpendicular distance of each chord PQ and RS from the centre is 6 cm. If the length of chord PQ is 18 cm, find the length of the other chord.

Both chords are at the same distance from the centre

Chords equidistant from the centre are equal

PQ = 18 cm $\implies$ RS = 18 cm

Answer: Length of the other chord = 18 cm

9. A line from the centre of a circle cuts a 10 cm chord at right angle where radius of the circle is 6 cm. What is the length from the centre to the chord?

The perpendicular from the centre bisects the chord

Half chord = $\frac{10}{2}$ = 5 cm, r = 6 cm

$d^2 = r^2 - (\text{half chord})^2 = 6^2 - 5^2 = 36 - 25 = 11$

$d = \sqrt{11} = 3.32$ cm

Answer: Length from the centre to the chord ≈ 3.32 cm

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10. In a circle, a perpendicular is drawn from the centre to chord AB. If mAB = 12 cm, what is the length of each segment after bisecting?

The perpendicular from the centre bisects the chord

Each segment = $\frac{1}{2}$ AB = $\frac{1}{2}$ × 12 = 6 cm

Answer: Each segment = 6 cm

Remember — Unit 8 formulas

Perpendicular from the centre bisects the chord: $r^2 = d^2 + (\text{half chord})^2$.

Equal chords ↔ equally far from the centre.

Equal arcs ↔ equal chords ↔ equal central angles.

One and only one circle passes through three non-collinear points.

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